

MINISTRY OF EARTH SCIENCES (MoES) — GOVERNMENT OF INDIA

NATIONAL INSTITUTE OF OCEAN TECHNOLOGY (NIOT, CHENNAI)
OPERATION NET-ZERO: TACTICAL RECOVERY DISPATCH ORDER

[DEMO MODE ACTIVE] This mission report contains simulated / demo acoustic survey data. Not for live maritime navigation.

Incident Ref ID:	INC-SHIPWRECK-002	Issue Timestamp:	2026-09-26 18:35:00 UTC
Dataset / Source:	AI4Shipwrecks (CC-BY-4.0)	AUV Asset:	NIOT Autonomous Surveyor-II

1. ACOUSTIC ANOMALY SPECIFICATION & MEASUREMENT PROVENANCE

Parameter	Value / Reading	Scientific Provenance
Anomaly Class:	SHIPWRECK-CLASS CONTACT	MEASURED (ONNX Tiled)
AI Confidence:	86.1%	MEASURED (Class Prob)
Target Elevation:	8.45 m	DERIVED (Trigonometry)
Seafloor Depth:	24.5 m	ASSUMED (Altimeter)
Coordinates:	Lat: 45.052100° N, Lon: -83.310400° E	DERIVED

2. MULTI-PING TEMPORAL TRACKING & CONTACT PERSISTENCE

Track Identifier:	TRK-WRECK-02	Persistence Status:	PERSISTENT
Observations:	6 consecutive pings	Track Age:	85.0 seconds elapsed

3. SYSTEM-1 EDGE REFLEX TELEMETRY (CONVAIINNOVATIONS/LAYA)

Reflex Decision:	REDUCE_SPEED
Hazard Severity Score:	6.8 / 10.0 (Deterministic Guardrail)
Execution Latency:	28.4 ms (Sub-35ms Non-Autoregressive)
Safety Disclaimer:	Advisory only — no direct vehicle control without operator authorization.

4. SYSTEM-2 TACTICAL REASONING BRIEFING (qwen/qwen3.8-27b (Groq Cloud) — ACTIVE)

Incident Summary:	Large metallic / timber hull structure detected on benthic floor with significant vertical acoustic relief.
Recovery Priority:	HIGH
Operator Directive:	Log coordinates for hydrographic heritage office; ascend 5 meters for stand-off pass.
Sensor Uncertainties:	- Structural integrity of mast protrusions uncertain - Archaeological status unverified

Generated By:
SAMUDRA-AI Autonomous Core
NIOT Acoustic Intelligence Unit

Verified By / Watch Officer:
Commander, Coast Guard Station Mandapam
Indian Coast Guard